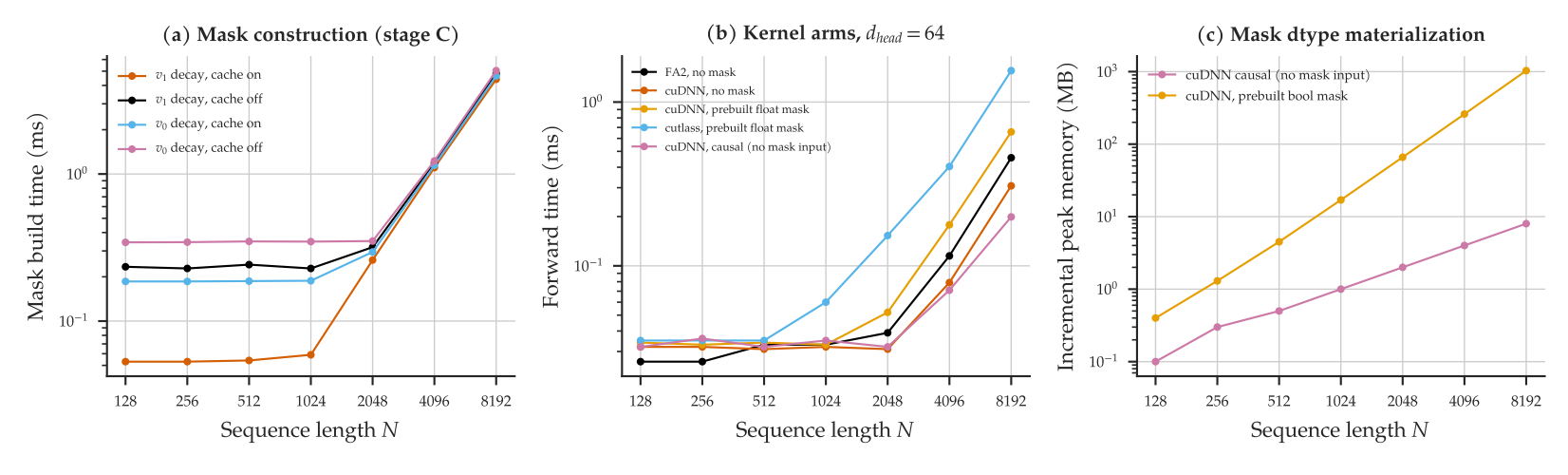
(a) Mask construction (stage C)
Mask build time (ms)
10^0
10^{-1}
v_1 decay, cache on
v_1 decay, cache off
v_0 decay, cache on
v_0 decay, cache off
128
256
512
1024
2048
4096
8192
Sequence length N
(b) Kernel arms, $d_{head} = 64$
Forward time (ms)
10^0
10^{-1}
FA2, no mask
cuDNN, no mask
cuDNN, prebuilt float mask
cutlass, prebuilt float mask
cuDNN, causal (no mask input)
128
256
512
1024
2048
4096
8192
Sequence length N
(c) Mask dtype materialization
Incremental peak memory (MB)
10^3
10^2
10^1
10^0
10^{-1}
cuDNN causal (no mask input)
cuDNN, prebuilt bool mask
128
256
512
1024
2048
4096
8192
Sequence length N